Proficient in Max for engineers (80 Hours)				
S.No	**Session**	**Topics Covered**	**Type**	**Duration (hrs)**
1	Session 1	**Introduction to Max** **Design Harmony** : Programming, Schematic design, Design development, Construction documentation, Contract administration, Max for Architects, Max for Mechanical Autodesk 3ds Max **Terminologies** : Modeling, Lights and Cameras, Animation, Rendering **The Welcome Screen** : Learn Tab, Start Tab, Extend Tab Layout Panel **3ds Max Interface**: Elements of GUI, Application button and Quick Access toolbar, Info Centre , Menu bar, Main toolbar, Ribbon, Scene Explorer, Viewport Layout, Command panel, Viewports, MAXScript Mini Listener, Status line and Prompt line, Isolate Selection toggle and Selection Lock toggle, Coordinate display, Animation and Time controls, Viewport Navigation controls **Viewport Navigation**: Go To End, Zoom, Zoom All, Zoom Region, Time Configuration, Field-of-View Button, Pan View, Orbit, Maximize Viewport Toggle Animation and Time Controls **Quad Menu** : Transform Quadrant, Display Quadrant **Tools Menu :** Display Floater, Array, Align, Snapshot, Color Clipboard Utility	Theory/Demo	2.0Hrs
2	Session 2	Practice 01: Set viewports and assign styles as shown.	Practical	2.0Hrs
3	Session 3	**Shapes and Geometry** **Spline**: Line, Rectangle, Circle, Ellipse Spline, Arc Spline, Donut Spline, NGon Spline, Star Spline, Helix Spline, Egg Spline, Section Spline, Extended Splines, WRectangle Spline, Channel Spline, Angle Spline, Tee Spline **Standard Primitives**: Box, Cone, Sphere, GeoSphere, Cylinder, Tube, Torus, Pyramid, Teapot, Plane, TextPlus **Extended Primitives**: Hedra, Torus Knot, Chamfer Box, ChamferCyl, Oil Tank, Capsule, Spindle, L-Ext, Gengon, C-Ext, Ring Wave, Prism, Hose, **Compound Objects:** Morph, Scatter, Conform, Connect, BlobMesh, Shape Merge, Boolean, Terrain, Loft, Mesher, ProBoolean, ProCutter **Windows:** Awning Window, Casement Window, Fixed Window, Pivoted Window, Projected Window, Sliding Window **AEC Extended:** Foliage, Railing, Wall **Doors:** Pivot Door, Sliding Door, BiFold Door **Stairs:** Straight Stair, L - Type Stair, U-Type Stair, Spiral Stair	Theory/Demo	2.0Hrs
4	Session 4	Practice 02: Model a single storey residence.	Practical	2.0Hrs
5	Session 5	**Shapes and Geometry** **Particle Systems**: Particle Flow (PF) Source, Spray, Snow, Super Spray, Blizzard, PArray, PCloud **Patch Grids**: Quad Patch, Tri Patch **Body Objects**: Body Utility Body Objects Join Bodies Body Cutter **NURBS Surfaces:** Point Surface , CV Surface Point Cloud Object	Theory/Demo	2.0Hrs
6	Session 6	Practice 03: Create and add particle system to scene.	Practical	2.0Hrs
7	Session 7	**Shapes and Geometry** **Dynamics Object:** Spring, Damper **Helpers**: Dummy, Container, Crowd, Delegate, Expose Transform (ExposeTM), Grid, Point, Light Meter, Tape, Protractor, Iray Section, Compass **Space Warp Objects:** Push, Motor Space Warp, Vortex, Drag, PBomb, Path Follow, Gravity, Wind, Displace **Hierarchy:** Adjust Pivot Rollout, Affect Pivot Only, Affect Object Only, Affect Hierarchy Only, Center to Object, Align to Object, Align to World, Center to Pivot, Align to Pivot, Align to World, Move Pivot to Origin, Reset Pivot **Motion :** Sub-Object, Parameters, Trajectories	Theory/Demo	2.0Hrs
8	Session 8	Practice 04: Modify geometry using specified space warp objects.	Practical	2.0Hrs
9	Session 9	**Modifiers** **Selection Modifiers:** Mesh Select Modifier, Patch Select modifier, Poly Select Modifier Editable Poly **Vertex:** Remove, Break, Extrude, Weld, Chamfer, Target Weld **Edge:** Remove, Split, Extrude, Weld, Chamfer, Target Weld, Bridge, Connect **Border:** Extrude, Insert Vertex, Chamfer, Cap, Bridge, Connect **Polygon/Element:** Insert Vertex, Extrude, Outline, Bevel, Inset, Bridge, Flip	Theory/Demo	2.0Hrs

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10	Session 10	Practice 05: Model bedroom furniture set.	Practical	2.0Hrs
11	Session 11	**Modifiers** **Object – Space Modifiers:** Bend Modifier, Lattice Modifier, Push Modifier, Shell Modifier, Stretch Modifier, Twist Modifier, Lathe Modifier, Wave Modifier, Smooth Modifier, TurboSmooth Modifier, Symmetry Modifier, Cloth Modifier, Simulation Parameters (Cloth) FFD (Free Form Deformation)	Theory/Demo	2.0Hrs
12	Session 12	Practice 06: Detail the bedroom furniture set using cloth and other modifiers.	Practical	2.0Hrs
13	Session 13	**Modelling Interior Objects** Detailed Modelling of a Kitchen Cabinet	Theory/Demo	2.0Hrs
14	Session 14	Practice 07: Model the kitchen cabinet.	Practical	2.0Hrs
15	Session 15	**Modelling Interior Objects** Modelling the Objects for Living Room	Theory/Demo	2.0Hrs
16	Session 16	Practice 08: Model Living room furniture set.	Practical	2.0Hrs
17	Session 17	**Modelling Exterior Objects** Modeling the Exterior of a Building	Theory/Demo	2.0Hrs
18	Session 18	Practice 09: Model the exterior of a building.	Practical	2.0Hrs
19	Session 19	**Modelling Exterior Objects** Environment Modeling	Theory/Demo	2.0Hrs
20	Session 20	Practice 10: Create an environment model as shown.	Practical	2.0Hrs
21	Session 21	**Modelling Mechanical Objects** Modeling a Headphone	Theory/Demo	2.0Hrs
22	Session 22	Practice 11: Model a headphone.	Practical	2.0Hrs
23	Session 23	**Modelling Mechanical Objects** Modeling a Gear System	Theory/Demo	2.0Hrs
24	Session 24	Practice 12: Model a gear system.	Practical	2.0Hrs
25	Session 25	**Materials** Material Editor **Compact Material Editor:** Get Material, Put Material to Scene ,Assign Material to Selection , Reset Map/Mtl to Default Settings, Make Material Copy, Make Unique, Put to Library, Material ID Channel flyout, Show Shaded/Realistic Material in Viewport flyout Show End Result, Go to Parent, Go Forward to Sibling	Theory/Demo	2.0Hrs
26	Session 26	Practice 13: Create and assign materials to the bedroom.	Practical	2.0Hrs
27	Session 27	**Materials** **Slate Material Editor:** Menu bar, Toolbar, Material/Map Browser, Status, Active View, View navigation, Parameter Editor **UVW Coordinates:** Coordinates Rollout (2D), Coordinates Rollout (3D), Camera Map Modifier (World Space) **UVW Map Modifier:** Manipulators for UVW Map	Theory/Demo	2.0Hrs
28	Session 28	Practice 14: Create and assign materials to the kitchen.	Practical	2.0Hrs
29	Session 29	**Materials** Composite Material Defining Material for Exterior Model Applying Material to Headphone Applying Metallic Materials	Theory/Demo	2.0Hrs
30	Session 30	Practice 15: Create and apply materials to the headphones.	Practical	2.0Hrs
31	Session 31	**Lights, Camera and Rendering** **Photometric Lights**: Target Light, Free Light, Mental Ray (mr) Skylight Portal, Sun Positioner **Standard Lights:** Target Spotlight, Free Spotlight, Target Directional Light, Free Directional Light, Omni Light, Skylight, mr Area Omni Light, mr Area Spotlight **Systems:** Sunlight and Daylight Systems, Sunlight Control Parameters rollout, Daylight Parameters rollout (Daylight system only) **Cameras**: Physical Camera, Free Camera, Target Camera, Walkthrough Assistant	Theory/Demo	2.0Hrs
32	Session 32	Practice 16: Make the model render ready- for interior render output.	Practical	2.0Hrs
33	Session 33	**Lights, Camera and Rendering** **Rendering**: Render Setup Dialog, Quicksilver Hardware Renderer, ART Renderer, Scanline Renderer, VUE File Renderer, iray Renderer, Mental ray Renderer, A360 Cloud Rendering, Environment and Effects Dialog Box Exposure Controls Exterior Lighting Environment Texturing Exterior Rendering	Theory/Demo	2.0Hrs
34	Session 34	Practice 17: Make the model render ready- exterior render output.	Practical	2.0Hrs

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35	Session 35	**Animation** Saving and Loading Animation, Animation Layers (Layer Controller), Enabling Animation Layers, Animation and Time Controls, Integrating Animation Layers in Workflow, Time Configuration **Animation Quad Menu:** Set quadrant, Pose quadrant, Transform quadrant	Theory/Demo	2.0Hrs
36	Session 36	Practice 18: Create an animated scene as shown.	Practical	2.0Hrs
37	Session 37	**Animation** Animation Constraints, Wire Parameters, Position/Rotation/Scale Controller, Video Post Day Light Shadow Animation Chess Animation Object on Path Simulation Gear Animation	Theory/Demo	2.0Hrs
38	Session 38	Practice 19: Create a daylight shadow animation and chess animation.	Practical	2.0Hrs
39	Session 39	**Files Management** **File – Handling:** New, Save, Merge, Manage, Properties, Exit Merging Files **AVI Files:** Compressor, Compression Quality, Keyframe Rate, Setup About the Print Size Wizard **Export:** OBJ Export, IGES Files, Exporting to 3DS Project Worksheet	Theory/Demo	2.0Hrs
40	Session 40	Practice 20: Create animation and get an AVI output. Export files in given extensions.	Practical	2.0Hrs